11-02 第2篇

阅读理解（说明文） Why Do Some Trees Lose Their Leaves in Autumn While Others Stay Green?

① Every November, a simple scene repeats itself in our city parks: the **gingko trees** on the south side of the playground turn a bright, soft yellow, and their leaves drift down into quiet piles on the stone path. At the same time, a few steps away, the **pine trees** by the east fence stand tall and green, as if winter had never sent a warning. Why do some trees, like gingkos and maples, drop their leaves every autumn, while others, like pines and firs, stay green all year round? The answer is both **clever** and **practical**. Trees, it seems, have been solving the puzzle of winter for millions of years.

② First, we should name the two teams. **Deciduous trees** (落叶树) are the trees whose leaves all fall together at one time each year — gingko, maple, poplar, Chinese parasol. **Evergreen trees** (常绿树) keep green leaves all year, although they actually drop older leaves slowly throughout the four seasons — pine, fir, cedar, camphor. The English word *deciduous* comes from a Latin root that means "to fall down". The word *evergreen* simply means "forever green".

③ The main problem in winter is **water**, not cold. When the ground freezes, roots cannot pull up as much water as before. Yet leaves, especially large flat ones, keep losing water through small holes on their surface, a slow and quiet process called **transpiration**. A tree with many big, thin leaves in winter is like a person wearing a soaking shirt on a freezing day: she will lose heat and water faster than she can replace them. So one good answer is to **drop the leaves** before winter and grow new ones in spring.

④ This is exactly what deciduous trees do. In autumn, the tree slowly closes off the thin bridge between the branch and the leaf, forming a small corky layer called an **abscission zone** (离层). Through this zone, the tree first **pulls back** valuable chemicals from the leaf into the branch — sugars, certain minerals, part of the green pigment **chlorophyll**. As the green fades, the yellow and red pigments that were there all along (hidden behind the green) suddenly become visible. That is why gingko leaves look yellow and maple leaves look red — not because a new colour has appeared, but because the old cover has been gently taken away.

⑤ Evergreens have solved the same winter problem in a very different way. First, their leaves are usually **thin needles**, much smaller than a broad flat leaf. A small surface means less water loss. Second, the needles are wrapped in a **waxy skin** that is like a thin raincoat; it keeps water inside and cold wind outside. Third, the **sugar liquid** inside pine

needles contains natural "anti-freeze" substances that stop ice crystals from forming. Because of these three defences, evergreens can keep their green leaves even when the thermometer drops below **−20 ℃** in some northern provinces.

⑥ There is also a big **size question**. A deciduous tree grows fast in spring and summer, pours all its energy into thin, flat leaves, and makes a lot of food (through photosynthesis) very quickly. An evergreen grows more slowly, but it never has to build a brand new set of leaves every year. A fir needle may live for **five to seven years** before it finally falls. In this way, evergreens are good for places with **short summers** — high mountains, cold north, windy sea-cliffs — where there is not enough warm time each year to grow a whole new leaf-set.

⑦ The two teams also **help** each other. Fallen deciduous leaves slowly break down on the forest floor, turning into rich soil from which evergreens — and many other plants — draw nutrients. In return, the dense green roof of evergreens gives small animals shelter in winter, keeping the same forest alive through the coldest months. A mixed forest, where gingkos and pines stand side by side, is not an accident. It is a **quiet partnership** formed over many thousands of years.

⑧ Some readers may ask: what about **bamboo**? Bamboo is green all year but is neither pine nor gingko. **Bamboo** is technically a **giant grass**, not a tree; its leaves have their own slow turnover, a little at a time. So in Chinese parks you can find **three** winter pictures: the bare gingko, the thick pine, and the bright green bamboo. Next time you walk through a park in late autumn, look up. Every tree above your head has written its own answer to the old, cold question — *how do I live through the winter?* The answer you see is hundreds of millions of years of careful editing, **quietly printed on every leaf**.

注释： deciduous 落叶的；evergreen 常绿的；transpiration 蒸腾作用；abscission zone 离层；chlorophyll 叶绿素；anti-freeze 抗冻的

根据短文内容，选择最佳答案。

1. What is the main problem trees face in winter, according to Paragraph ③?

A. Heavy snow crushing the leaves.
B. Not enough sunlight.
C. Water: frozen ground stops roots pulling up enough water, while big flat leaves keep losing water through transpiration.
D. Too many insects.

2. Why do gingko leaves turn yellow and maple leaves turn red in autumn?

A. Because a new colour grows in their leaves in October.
B. Because the tree slowly pulls back chlorophyll through the abscission zone, and the yellow and red pigments (hidden behind the green all along) become visible.

C. Because wind paints the leaves.
D. Because the trees die every year.

3. Which THREE defences help evergreens keep their leaves through the cold?

A. Thin needles, a waxy skin, and natural "anti-freeze" sugar liquid inside the needles.
B. Big flat leaves, red colour, and plenty of water.
C. Very large size, strong roots, and noisy wind.
D. Fallen leaves, bark, and dead branches.

4. According to Paragraph ⑥, why are evergreens good for places with SHORT summers?

A. Because they grow very big very quickly.
B. Because a fir needle may live for five to seven years before it falls, so evergreens do not need enough warm time each year to build a whole new leaf-set.
C. Because they do not need sunlight.
D. Because they like very hot places.

5. What is the special fact about bamboo in Paragraph ⑧?

A. Bamboo is a kind of pine tree.
B. Bamboo is technically a giant grass, not a tree; its leaves have their own slow turnover, a little at a time.
C. Bamboo leaves all fall in November.
D. Bamboo only grows in the north.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第③段 "water, not cold ... transpiration"
【解析】水分流失为核心难题。A、B、D 均错。故选 C。

2.
【答案】B
【题型】细节理解题
【定位】第④段 "pulls back ... chlorophyll ... yellow and red pigments ... hidden behind the green ... become visible"
【解析】原色显现原理。A、C、D 均错。故选 B。

3.
【答案】A
【题型】细节理解题

【定位】第⑤段三防御
【解析】针状 + 蜡面 + 抗冻糖液。B、C、D 均错。故选 A。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段 "fir needle may live for five to seven years ... short summers"
【解析】不需每年重建叶片。A、C、D 均错。故选 B。

5.
【答案】B
【题型】细节理解题
【定位】第⑧段 "Bamboo is technically a giant grass ... slow turnover, a little at a time"
【解析】分类 + 换叶特性。A、C、D 均错。故选 B。